

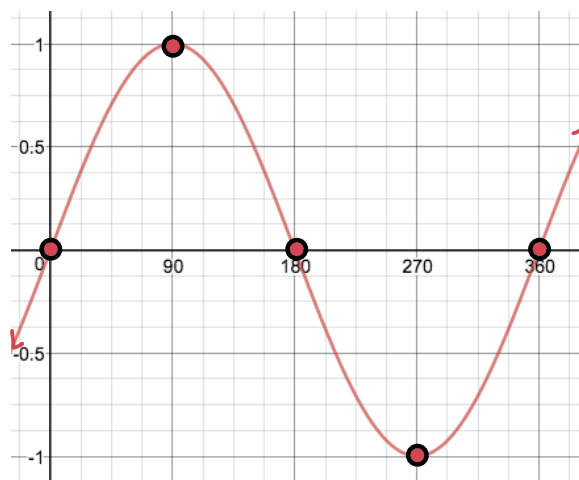
Section 1: Review of Sine and Cosine Functions

$$y = a \sin[k(x - d)] + c \text{ OR } y = a \cos[k(x - d)] + c$$

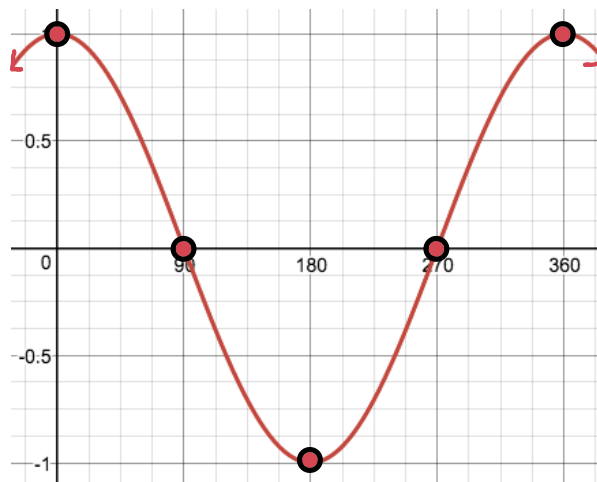
a	k	d	c
Vertical stretch or compression by a factor of a . Vertical reflection if $a < 0$ $ a = \text{amplitude}$	Horizontal stretch or compression by a factor of $\frac{1}{k}$. Horizontal reflection if $k < 0$. $\frac{360}{ k } = \text{period}$	Phase shift $d > 0$; shift right $d < 0$; shift left	Vertical shift $c > 0$; shift up $c < 0$; shift down

Graphs of parent functions $y = \sin x$ and $y = \cos x$ using key points:

x	y
0	0
90	1
180	0
270	-1
360	0



x	y
0	1
90	0
180	-1
270	0
360	1



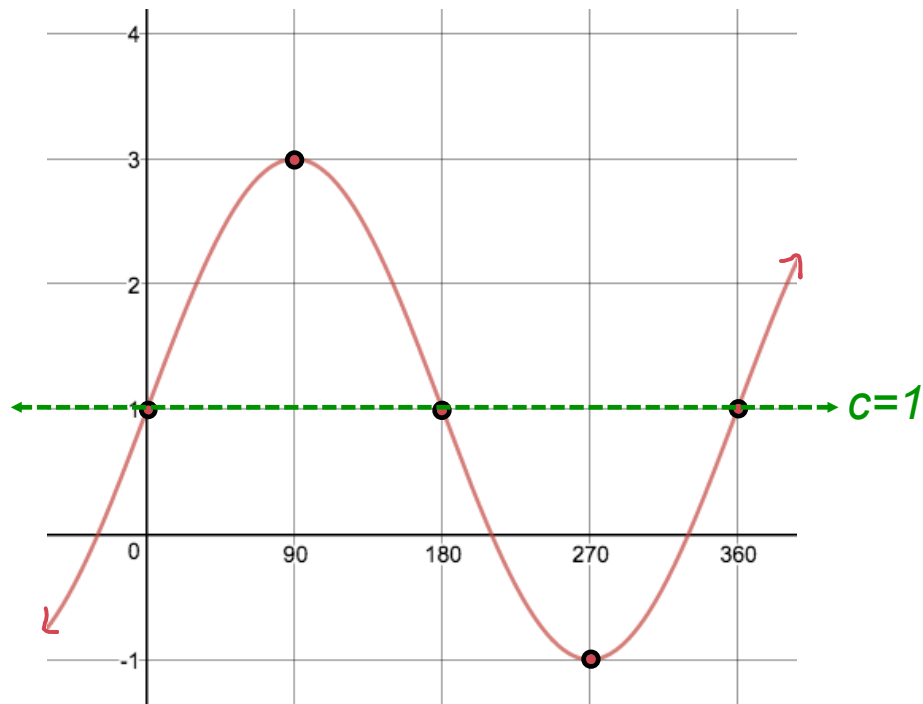
Section 2: Graphing Transformed Sinusoidal Functions

Example 1: Graph $y = 2 \sin x + 1$ using transformations. Then state the amplitude, period, and number of cycles between 0° and 360° .

$a = 2$; vertical stretch by a factor of 2 ($2y$)
 $c = 1$; vertical shift up 1 unit ($y + 1$)

$y = \sin x$	
x	y
0	0
90	1
180	0
270	-1
360	0

$y = 2 \sin x + 1$	
x	$2y + 1$
0	1
90	3
180	1
270	-1
360	1



Amplitude: $\text{amplitude} = |a| = 2$

Period: $\text{period} = \frac{360}{|k|} = \frac{360}{1} = 360^\circ$

Number of cycles between 0° and 360° : $\text{\# of cycles} = |k| = 1$

Example 2: Graph $y = -1.5 \cos[3(x - 30^\circ)] + 0.5$ using transformations. Then state the amplitude, period, and number of cycles between 0° and 360° .

$a = -1.5$; vertical stretch by a factor of 1.5 and a vertical reflection ($-1.5y$)

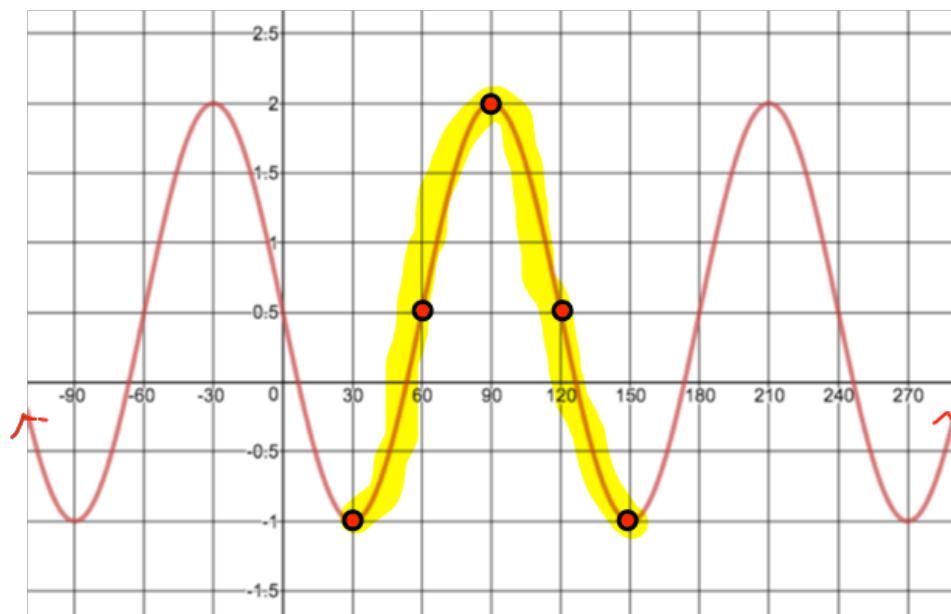
$k = 3$; horizontal compression by a factor of $\frac{1}{3}$ ($\frac{x}{3}$)

$d = 30$; phase shift 30° to the right ($x + 30$)

$c = 0.5$; vertical shift 0.5 units up ($y + 0.5$)

$y = \cos x$	
x	y
0	1
90	0
180	-1
270	0
360	1

$y = -1.5 \cos[3(x - 30^\circ)] + 0.5$	
$\frac{x}{3} + 30$	$-1.5y + 0.5$
30	-1
60	0.5
90	2
120	0.5
150	-1



Amplitude: $\text{amplitude} = |a| = 1.5$

Period: $\text{period} = \frac{360}{|k|} = \frac{360}{3} = 120^\circ$

Number of cycles between 0° and 360° : $\text{\# of cycles} = |k| = 3$

Example 3: Graph $y = \sin[-4(x - 60^\circ)] + 2$ using transformations. Then state the amplitude, period, and number of cycles between 0° and 360° .

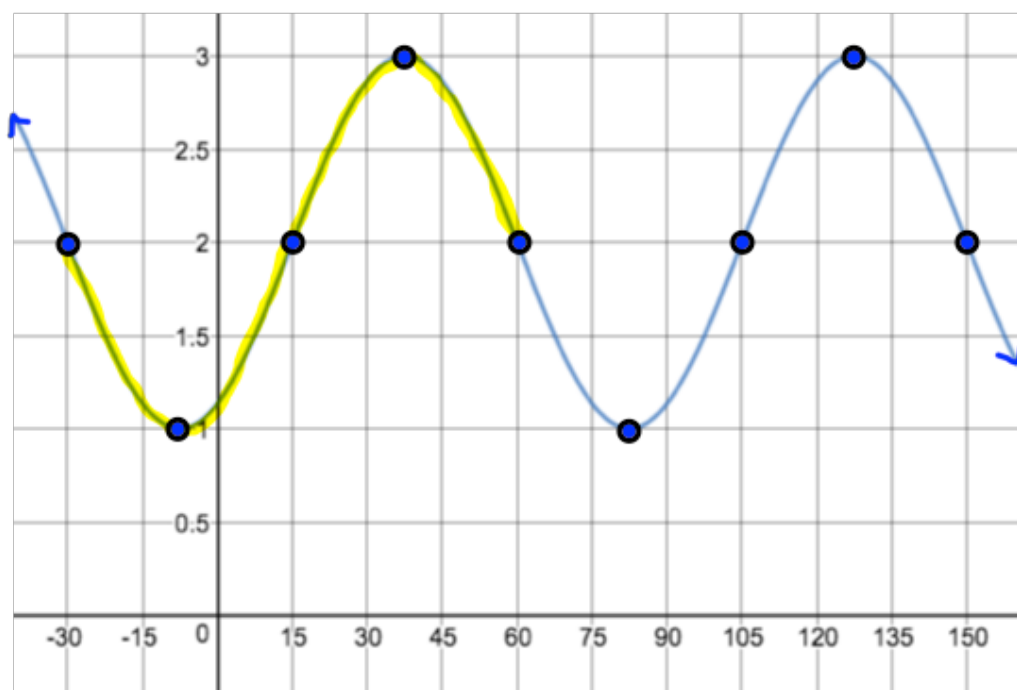
$k = -4$; horizontal compression by a factor of $\frac{1}{4}$, and horizontal reflection ($-\frac{x}{4}$)

$d = 60$; phase shift 60° to the right ($x + 60$)

$c = 2$; vertical shift 2 units up ($y + 2$)

$y = \sin x$	
x	y
0	0
90	1
180	0
270	-1
360	0

$y = \sin[-4(x - 60^\circ)] + 2$	
$-\frac{x}{4} + 60$	$y + 2$
60	2
37.5	3
15	2
-7.5	1
-30	2



Amplitude: $amplitude = |a| = 1$

Period: $period = \frac{360}{|k|} = \frac{360}{4} = 90^\circ$

Number of cycles between 0° and 360° : $\#of\ cycles = |k| = 4$